# Project Report: Global Groundwater Depletion Analysis (1990–2023)

*Spatio-Temporal Assessment of Water Stress and Aquifer Decline Across Asia and Beyond.*

Author: Mulagundla Sujith Reddy
Affiliation: Malla Reddy University, Hyderabad

## 1. Executive Summary

This project investigates the accelerating depletion of groundwater resources across 32 countries between 1990 and 2023, focusing on Asia—the most water-stressed continent. Using real-world datasets from the World Bank, NASA's GRACE mission, and FAO AQUASTAT, the study identifies critical depletion zones and quantifies the relationships between groundwater stress, population growth, and agricultural demand. The results reveal alarming declines in aquifer levels exceeding 30% in several regions, particularly in the Middle East and South Asia. The findings provide a quantitative, data-driven basis for immediate policy intervention, sustainable management, and international cooperation.

## 2. Introduction & Objectives

Groundwater supports nearly 2 billion people globally and represents about 30% of the planet's freshwater. However, unsustainable extraction has led to serious declines, land subsidence, and ecosystem damage. This study aims to quantify and visualize groundwater depletion patterns across continents, analyze trends, and identify socio-economic and climatic drivers influencing water stress.

**Objectives:**

- Quantify regional patterns of groundwater depletion (1990–2023).
- Analyze temporal trends in water stress and aquifer decline.
- Identify correlations with socio-economic and climatic variables.
- Provide actionable recommendations for sustainable water management.

## 3. Methodology & Technical Approach

**Technology:** Python (Pandas, NumPy, Matplotlib, Plotly, wbdata, SciPy) for data analysis and visualization.

**Data Sources:** Integrated datasets from World Bank Open Data API, NASA GRACE Mission, FAO AQUASTAT, and peer-reviewed scientific studies. The pipeline involved data cleaning, integration, index computation, correlation analysis, and geospatial visualization.

**Pipeline Overview:**

1. Data Acquisition via World Bank API for indicators such as water stress, population, and withdrawals.
2. Integration with GRACE satellite-based depletion rates.
3. Derived groundwater index (GWI) calculation.
4. Correlation and temporal analysis.
5. Visualization through interactive maps and charts.

## 4. Data Overview

**Geographic Scope:** 32 countries across Asia, Middle East, Africa, and Americas.
**Time Period:** 1990–2023
**Key Indicators:** Water stress (%), groundwater depletion rate (m/year), freshwater withdrawals, population, agricultural land use, and precipitation.
**Average Water Stress (2023):** 67.3%
**Average Groundwater Level Decline:** 32.7% since 1990.

## 5. Results & Key Findings

- The Middle East and South Asia emerge as global hotspots, with stress levels exceeding 100% in seven countries.
- UAE and Saudi Arabia show annual depletion rates of 1.2–1.3 m/year.
- India and Pakistan exhibit 25–30% declines in groundwater levels, threatening agricultural sustainability.
- Correlation analysis shows strong linkage between water stress and withdrawals ($r = 0.71$), and a strong negative correlation between stress and groundwater levels ($r = -0.82$).

## 6. Recommendations

**Immediate Actions (0–2 Years):**

- Declare groundwater emergency zones in top 10 stressed countries.
- Deploy 10,000+ automated monitoring stations and satellite-based sensors.
- Pilot subsidy reforms replacing free electricity with volumetric pricing.

**Medium-Term Reforms (2–5 Years):**

- Enact groundwater regulation laws and enforce extraction limits.
- Invest in recharge infrastructure to add 500 billion $m^3$/year of storage capacity.
- Promote crop shifts to less water-intensive varieties.

**Long-Term Transformations (5–15 Years):**

- Implement full-cost water pricing for all sectors.
- Achieve 80% wastewater reuse through decentralized treatment.
- Develop climate-resilient groundwater management and aquifer recovery programs.

## 7. Conclusion

This project demonstrates that groundwater depletion is an accelerating global crisis, tightly linked to population growth, agricultural demand, and weak governance. At current depletion rates, regions like South Asia and the Middle East may reach critical scarcity within 20 years. However, nations such as Israel and Singapore illustrate that with data-driven policy, efficient water reuse, and strict regulation, sustainable groundwater management is achievable.